

Good evening, everyone. We are SpectraLink, a team from Peking University and Beijing Union University. Our project is called **SpectraLink: A Neuro-Symbolic AI Support System for Caregivers of Children with Autism Spectrum Disorder**.

First, please look at the **left side of our poster**, where we briefly introduce the background and the evidence base.

According to the Centers for Disease Control and Prevention, ASD prevalence among 8-year-old children increased from 1 in 36 in 2020 to 1 in 31 in 2022, while many children require lifelong care. This places a substantial and prolonged burden on families, while caregivers often have limited access to timely and evidence-based support in home.

To address their difficulty, we are developing **SpectraLink, an app for caregivers**. The app combines a large language model with a knowledge graph containing over 40,000 structured triples from evidence-based sources. Based on 36 caregiver interviews, we identified key needs for personalized guidance, mental health support, practical resources, and easy-to-use tools.

Now, please turn to the **center of the poster**, which presents the main functions of our app.

The first is the **home screen**.

The second, and most important, is **caregiving guidance**. As shown in our demonstration, the app recognizes caregivers' situation, retrieves relevant

evidence from the knowledge graph, and provides practical, step-by-step suggestions that can be applied in daily caregiving.

The other functions include **caregiver mental health support**, a **community section** for caregivers to communicate and share information, and a **help center** containing relaxation tools, professional resources, and links to relevant services.

Finally, please look at the **right side of the poster**, where we present our business model.

In **Phase One**, we will conduct institutional pilots in ASD rehabilitation centers, which can recommend SpectraLink to caregivers as an after-session support tool, extending professional support from rehabilitation settings into the home.

In **Phase Two**, we will introduce a three-tier caregiver-facing model. The basic tier will provide free, knowledge-graph-grounded caregiving guidance. The intermediate tier will offer group consultation sessions, while the premium tier will provide one-to-one expert consultations.

At the same time, we are currently exploring potential collaboration with **schools serving children with autism, artificial intelligence companies, and hospitals**. These partnerships may support pilot testing, technical development, professional consultation, and future implementation.

Finally, by delivering guidance digitally and reducing reliance on printed

materials and unnecessary travel for routine consultations, the project may help reduce resource consumption and travel-related emissions, supporting a more sustainable long-term care model.

Through SpectraLink, we hope to connect evidence-based professional knowledge with caregivers' everyday needs and provide support that is accessible, reliable, practical, and sustainable.

Thank you.

Good evening, everyone. We are SpectraLink from Peking University and Beijing Union University, and our project is a neuro-symbolic AI support system for caregivers of children with autism spectrum disorder. 15

Leftside showed the background. According to the Centers for Disease Control and Prevention, ASD prevalence among 8-year-olds has risen from 1 in 36 in 2020 to 1 in 31 in 2022. Many children need lifelong care, which places a heavy and lasting burden on families, yet caregivers often lack timely, evidence-based support at home. 40

To address this difficulty, we are developing SpectraLink—an app that combines a large language model with a knowledge graph containing over 40,000 evidence-based triples. Based on 36 caregiver interviews, we identified key needs for personalized guidance, mental health support, practical resources, and easy-to-use tools. 1'10

The most important feature is caregiving guidance. As shown in the middle of our poster, the app recognizes the caregiver's situation, retrieves relevant evidence from the knowledge graph, and provides step-by-step suggestions. Other features include a community section for sharing information, caregiver mental health support, and a help center with relaxation tools, professional resources, and service links. 1'40

Regarding our business model, please come to the right side. In phase one we will run institutional pilots at ASD rehabilitation centers, where the app can be

an after-session support tool, extending professional help from the center into the home. In phase two, we will introduce a three-tier caregiver-facing model: a basic tier with free, knowledge-based guidance; an intermediate tier with group consultations; and a premium tier offering one-to-one expert advice. 2'15

At the same time, we are exploring partnerships with schools for autistic children, AI companies, and hospitals to support future implementation. 2'25

Finally, by delivering guidance digitally, we reduce reliance on unnecessary travel for consultations, which may help lower travel-related emissions, and support a more sustainable care model. 2'40

To sum up, through SpectraLink, we hope to connect evidence-based knowledge with caregivers' everyday needs, and provide support that is accessible, reliable, and sustainable. Thank you. 3'